

# Computational chemist playbook: clinic-viable BACE1

BACE1\_Discovery canvas (print of all tabs)

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Your job is not another catalytic IC50. It is to find a ligand that lowers APP  $\beta$ -cleavage 20–40% while leaving SEZ6/NRG1/CHL1/gp130 intact, with CNS-like ADMET. Most published BACE1 docking papers optimize the binding mode that already failed Phase 3.

**Do not start here:** virtual screening of the OM99-2 / Asp32–Asp228 site, FEP on verubecestat analogs, or a generative model trained on ChEMBL BACE1 IC50.

## 1. Project portfolio

| ID   | Project                     | Priority  | Core computation                                                                               | Success look                                                             |
|------|-----------------------------|-----------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| P0-A | Substrate atlas             | Do first  | AF3/Boltz + MD of BACE1 with APP WT/Swe/A673T/A673V vs SEZ6, NRG1, CHL1, gp130 at pH 4.5       | APP-unique contacts and Waters SEZ6 does not use                         |
| P0-B | Icelandic mimic             | Do first  | QM/MM and FEP on P2' Ala vs Thr vs Val; favor $\beta'$ (Glu11) over $\beta$ (Asp1) on APP only | Shift C99→C89 in silico; SEZ6 geometry unchanged                         |
| P0-C | APP-BACE1 PPI / ASBI screen | Do first  | Hotspots on APP juxta + BACE1 loops; VS for dual binders (FAH65-like)                          | Docked to APP or encounter complex; displaced by APP peptide, not OM99-2 |
| P1-D | Exosite small molecules     | Second    | 3R1G Fab epitope; VS + MD allostery                                                            | Flap/10s motion that hurts APP more than SEZ6 — must prove               |
| P1-E | Anti-target filters         | Always on | BACE2, CatD, pepsin, renin; P-gp; CNS MPO                                                      | CatD inactive; BACE2 $\gg$ 30 $\times$ ; not efflux                      |
| P1-F | Low-dose catalytic backup   | Optional  | Occupancy-PD for 30% A $\beta$ , not 80%                                                       | Only if P0 fails and sSEZ6 $\geq$ 70% at that occupancy                  |
| P2-G | Membrane encounter          | Later     | CG then AT MD, endosome-like bilayer, pH 4.5                                                   | Poses that block APP recruitment, not the scissile bond                  |

| ID   | Project                      | Priority         | Core computation                           | Success look                                                               |
|------|------------------------------|------------------|--------------------------------------------|----------------------------------------------------------------------------|
| P2-H | Generative on the wrong site | Avoid as primary | REINVENT/diffusion on ChEMBL BACE1 actives | Will reinvent verubecestat unless APP-selective reward + catalytic penalty |

This week: curate 1FKN / 3R1G / flap-open apo / BACE2 / CatD / APP C99; Asp32/Asp228 at pH 4.5; 8-15-mer peptides; AF3 complexes; dock clinical catalytics as negative fingerprints; fpocket/FTMap tagged catalytic vs exosite vs PPI.

## 2. Master workflow (four filters)

1. **Binding-mode.** Pose on APP, APP-BACE1 interface, or 3R1G exosite. If it overlaps OM99-2 / catalytic water / Asp dyad, it is a low-dose backup. Checks: RMSD to 1FKN ligand > 3 Å; no H-bond to both Asp32 and Asp228; noncompetitive vs OM99-2.
2. **Substrate-selectivity.** MM-GBSA / FEP / PELE on BACE1-APP vs BACE1-SEZ6/NRG1. Target  $\Delta\Delta G$  (APP vs SEZ6)  $\geq 2$  kcal/mol. If both blocked, it is verubecestat with extra steps.
3. **Anti-target.** Same pose family into BACE2 and CatD. Retinal CatD killed LY2811376. Pass: predicted CatD  $K_i > 10 \mu M$ ; BACE1/BACE2  $\geq 30\times$ .
4. **CNS developability.** CNS MPO  $\geq 4$ , MW < 450, TPSA 40-90, cLogP 2-4, HBD  $\leq 2$ , P-gp non-substrate. Also hERG, CYP3A4 TDI, hepatotox alerts (atabecestat/LY2886721/RG7129).

**Method stack.** AF3/Boltz/HADDOCK; fpocket/FTMap/SiteMap; GNINA/rDock/Glide ensemble (flap-open and closed); OpenMM/GROMACS 3x500 ns pH 4.5; ABFE/RBFE on APP vs SEZ6; QM/MM gem-diol for WT/A673T/Swe/SEZ6; ML must penalize ChEMBL BACE1 actives; ADMET before buying.

BACE1 works in endosomes. Dyad protonation at pH 4.5-5.0  $\neq$  pH 7.4. Report interface screens at both.

## 3. Structures and models

| System                  | PDB / source                                         | Role                     |
|-------------------------|------------------------------------------------------|--------------------------|
| BACE1 + OM99-2          | 1FKN                                                 | Anti-target fingerprint  |
| Clinical catalytics     | verubecestat, lanabecestat, elenbecestat co-crystals | Negative controls        |
| Apo flap open/closed    | 1W50 + MD ensemble                                   | Ensemble docking         |
| Exosite Fab             | 3R1G                                                 | Allosteric pharmacophore |
| BACE2                   | e.g. 2EWY                                            | PMEL / hair anti-target  |
| Cathepsin D             | 1LYB                                                 | Retinal tox hard filter  |
| APP C99 / $\beta$ -site | 2LP1 NMR; WT, Swe, A673T, A673V                      | On-target                |
| SEZ6, NRG1, CHL1, gp130 | AF3 cleavage-site models                             | Safety substrates        |

Registers: APP WT ...VKM↓DAEFRHDS...; Swedish ...VNL↓DAEF... (~60x better — do not overfit); A673T ...VKM↓DTEF...; A673V pathogenic contrast;  $\beta'$  ...DAEFRH↓DSGY...; SEZ6 motif must not have a raised TS barrier.

FAH65 has no public co-crystal. Treat as SAR seed; require dual recognition. If analogs collapse into the Asp dyad, the series is not APP-selective.

## 4. Comp to wet lab

| Your output                        | Wet experiment                              | Kill if                     |
|------------------------------------|---------------------------------------------|-----------------------------|
| ASBI / PPI shortlist               | BACE1 × MBP-APP-C125 vs SEZ6 / NRG1 / PSGL1 | SEZ6 IC50 within 10× of APP |
| Predicted noncompetitive vs OM99-2 | SPR vs APP and BACE1; OM99-2 competition    | Competitive with OM99-2     |
| pH 4.5 pose                        | Neurons: sAPPβ, Aβ, sSEZ6, sCHL1, sgp130    | sSEZ6 falls with sAPPβ      |
| CNS MPO / P-gp pass                | MDCK-MDR1, mouse Kp,uu                      | Efflux or no free brain     |
| CatD reject list                   | CatD biochemical ± retinal explant          | LY2811376-like CatD         |
| 30% Aβ occupancy model             | 7-day mouse PD                              | Need 70% Aβ drop to see PD  |
| Spine-safe prediction              | 21-day two-photon at 20–40% Aβ              | Spine loss (sSEZ6 ~17%)     |

If you only run three campaigns: (1) APP vs SEZ6/NGR1 atlas at pH 4.5; (2) VS of APP-interface + 3R1G with catalytic poses auto-rejected; (3) FEP selectivity + CNS/CatD/BACE2, then 20 compounds to the neuronal panel.

## Sources

Hong 2000 *Science* (1FKN); Patel 2004 *J Mol Biol* (1W50); Atwal 2011 *Sci Transl Med* (3R1G); Barrett 2012 *Science* (2LP1). May 2011 / Zuhl 2016 (CatD). Rochin 2013 (BACE2/PMEL). Jonsson 2012; Pignoni 2016; Willem 2006; Kuhn 2012; Pratsch 2023. Campagna 2025 (FAH65). Wager 2010 (CNS MPO). Tian 2020 (ff19SB). Olsson 2011 (PROPKA3). Egan 2018/2019; Wessels 2020 (clinical negative controls).